CSCI 1810 10/29

Context: The Knuth-Morris-Pratt algorithm finds the first instance of a pattern p in a text t via a DFA. We have a skeletal DFA M_p, where each state j indicates having read the longest prefix $p_1p_2 \dots p_j$ of p that is a suffix of $t_1t_2 \dots t_k$. But we are missing transitions for the case $t_{k+1} \neq p_{j+1}$.

Definition of the Failure Function

When $t_{k+1} \neq p_{j+1}$, M_p should transition to the highest numbered state i such that $p_1p_2 \dots p_i$ is a suffix of $t_1t_2 \dots t_{k+1}$. To easily identify such states, we will construct a failure function.

The failure function f of a pattern p is defined so that $f(j)$ is the highest number s such that $s < j$ and $p_1p_2 \dots p_s$ is a suffix of $p_1p_2 \dots p_j$, or 0 if s does not exist. For example, if $p = aabbaab$, then $f(1) = 0, f(2) = 1, f(3) = 0, f(4) = 0, f(5) = 1, f(6) = 2$, and $f(7) = 3$.

Application of the Failure Function

When $t_{k+1} \neq p_{j+1}$, we recursively apply f m times, where m is the smallest number such that

1. $f^{(m)}(j) = u$ and $t_{k+1} = p_{u+1}$, or
2. $f^{(m)}(j) = 0$ and $t_{k+1} \neq p_1$.

Note on notation: $f^{(m)}(j) = \underbrace{f(f(\dots f}_{m \text{ times}}(j)\dots))$

In Case 1, M_p transitions to state $u + 1$, and in Case 2, M_p transitions to state 0. Observe that M_p terminates at the accepting state ℓ, where ℓ is the length of p, if and only if p occurs in t.

Computation of the Failure Function

The failure function can be computed for all j in linear time using the following algorithm:

$f(1) = 0$
$i = 0$
`For` j `in` $2, 3, \dots, \ell$`:`
    $i = f(j-1)$
    `While` $p_j \neq p_{i+1}$ `and` $i > 0$`:`
        $i = f(i)$
    `If` $p_j \neq p_{i+1}$ `and` $i = 0$`:`
        $f(j) = 0$
    `Else:`
        $f(j) = i + 1$

For next time: After applying the failure function, M_p seemingly needs nondeterminism to transition without consuming anything from the input. However, we will see that determinism is sufficient.